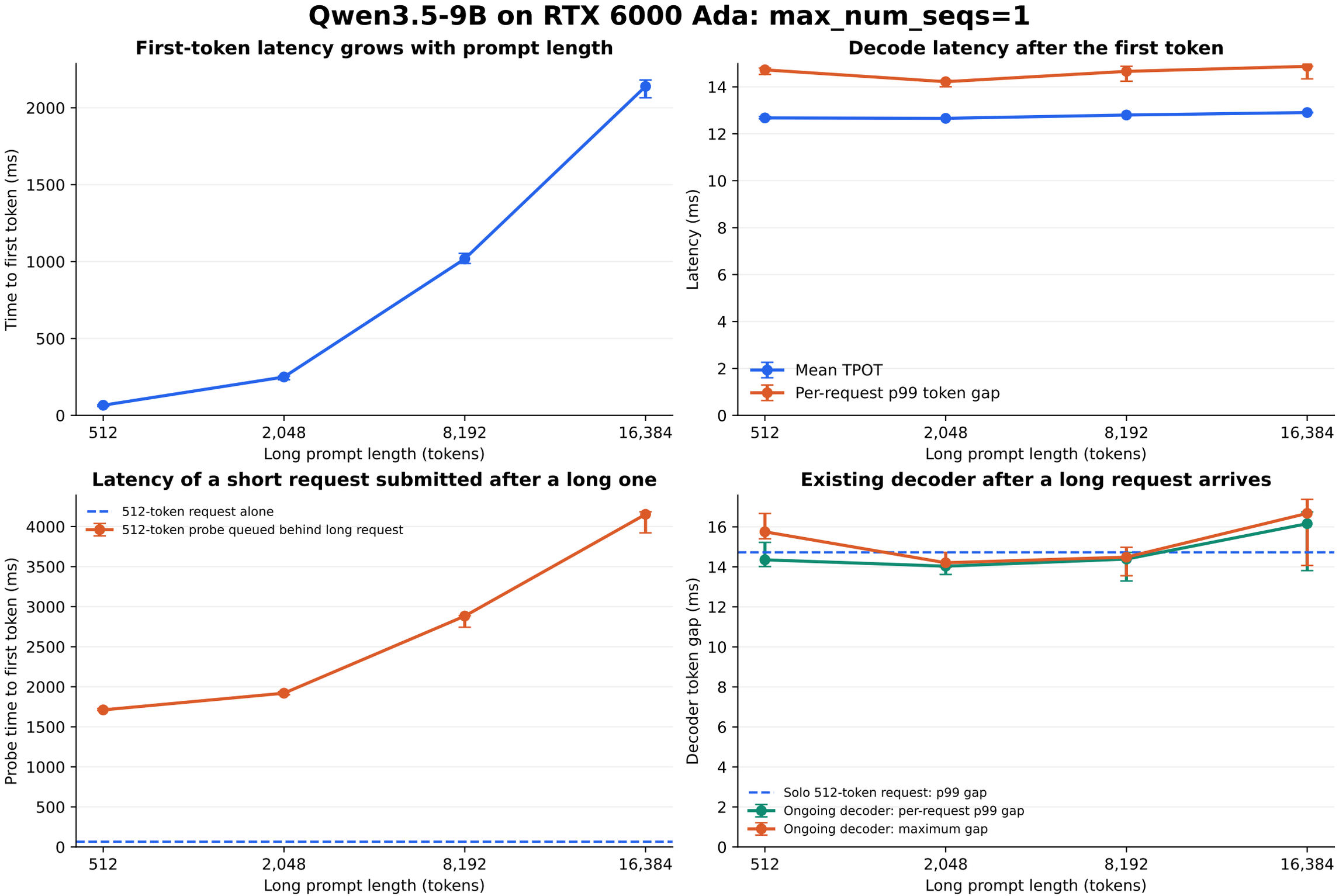
Qwen3.5-9B on RTX 6000 Ada: max_num_seqs=1
First-token latency grows with prompt length
Time to first token (ms)
0
500
1000
1500
2000
512
2,048
8,192
16,384
Long prompt length (tokens)
Decode latency after the first token
Latency (ms)
0
2
4
6
8
10
12
14
Mean TPOT
Per-request p99 token gap
512
2,048
8,192
16,384
Long prompt length (tokens)
Latency of a short request submitted after a long one
Probe time to first token (ms)
0
500
1000
1500
2000
2500
3000
3500
4000
512-token request alone
512-token probe queued behind long request
512
2,048
8,192
16,384
Long prompt length (tokens)
Existing decoder after a long request arrives
Decoder token gap (ms)
0
2
4
6
8
10
12
14
16
Solo 512-token request: p99 gap
Ongoing decoder: per-request p99 gap
Ongoing decoder: maximum gap
512
2,048
8,192
16,384
Long prompt length (tokens)
GPTQ INT4 weights · FP8 KV · eager · max_num_seqs=1 · 3 repetitions · median with min–max bars
Cold-prefix requests; loopback client timings. This measures queueing, not concurrent prefill/decode interference.